AALBORG UNIVERSITY
DENMARK

12 RESEARCH PAPER & COURSE OVERVIEW

HEVNER, A. & GREGOR, S. (2022). ENVISIONING ENTREPRENEURSHIP AND DIGITAL INNOVATION THROUGH A DESIGN SCIENCE RESEARCH LENS: A MATRIX APPROACH. INFORMATION & MANAGEMENT, 59(3), 103350.

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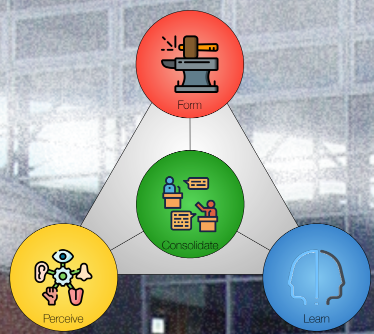
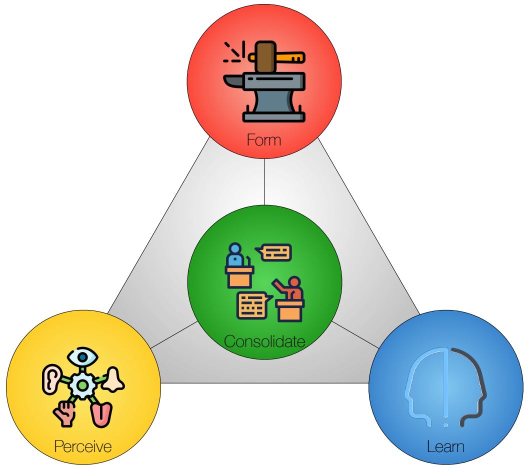


Agenda

- Exercise 11 Follow-up.
- Research Paper:
  Envisioning entrepreneurship and digital innovation through a design science research lens: A matrix approach.
- Course Overview (Separate file).

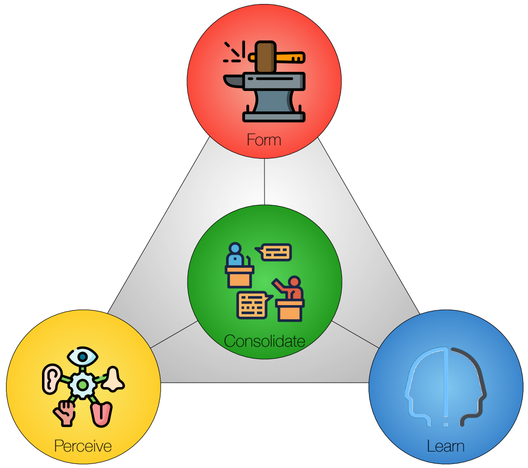


Follow-up: Exercise 11

Exercise 11 (Merit)

This exercise concerns functional and non-functional requirements, as well as strategies for improving robustness and managing dependencies.

- Discuss the functional completeness of your design and state *reservations*, *rebuttals*, and *values* accordingly.
- Discuss the robustness of your design and assess if any of the strategies in section 18.2 would be helpful.
- Outline dependencies in your design and assess the utility of the strategies in section 18.3.
- Use ETVX (section 18.4) to model robustness and dependencies.



Research Paper

Hevner, A. & Gregor, S. (2022). Envisioning entrepreneurship and digital innovation through a design science research lens: A matrix approach. *Information & Management,* 59(3), 103350.

Why they wrote the essay

- *Design Science Research (DSR) in the information systems (IS) field is, at its essence, about Digital Innovation (DI).*
- *Innovative sociotechnical design artifacts involve digital information technologies (IT) being used in ways that result in profound disruptions to traditional ways of doing business and to widespread societal changes.*
- *In this paper we present a new matrix approach to DI based on DSR, entrepreneurship, and innovation theories.*
- *We define the combined DSR and DI matrix approach in terms of four strategies: invention; advancement; exaptation; and exploitation and their associated DI practices.*

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Main Concepts

- *Digital Innovation:* The appropriation of digital technologies in the process of and as the result of innovation. Digital innovation emphasizes combinations of digital and physical components to produce novel products.
- *Design Science Research:* DSR in the study of sociotechnical information systems creates and evaluates IT artifacts to solve organizational problems via a rigorous research focus that requires the contribution of new, transferable knowledge not known to previously exist.

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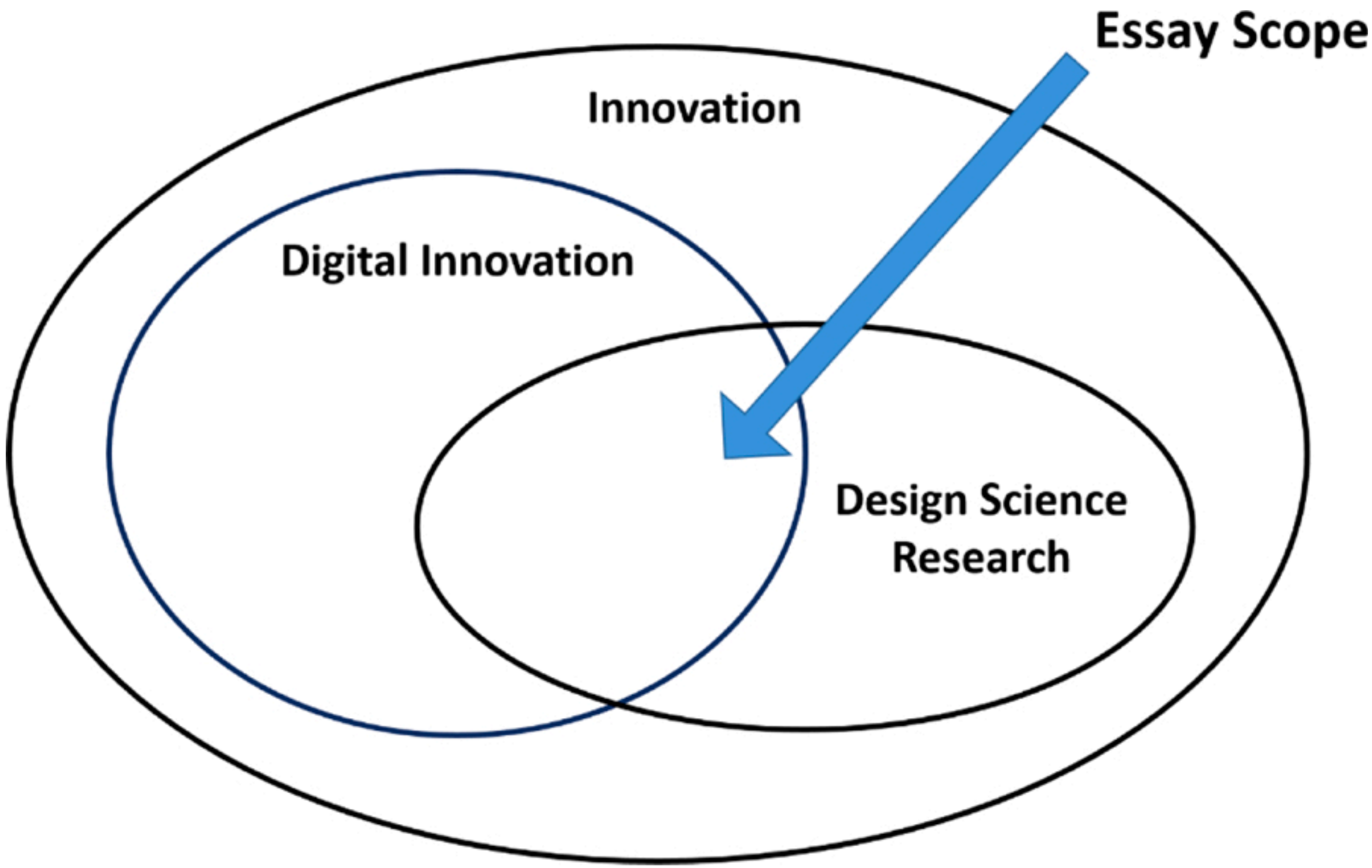
Scope of Essay – Digital Innovation and Design Science Research



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Main Approach: Effectuation

- *Effectuation processes take a set of means as given and focus on selecting between possible effects that can be created with that set of means. Causation processes take a particular effect as given and focus on selecting between means to create that effect.*
- *Effectuation logic has some commonality with the idea of "need-solution" pairs advanced by von Hippel and von Krogh, who argue that problem formulation need not precede problem solving, but rather a need and a solution may be discovered together.*

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Distinguishing Characteristic

The distinguishing characteristic between causation and effectuation is in the set of choices:

- *Choosing between*
  - ***means** to create a particular effect (causation),*
  - ***many possible effects** using a particular set of means (effectuation)*
- *Causation models consist of **many-to-one** mappings*
- *Effectuation models involve **one-to-many** mappings*

Sarasvathy, S. D.: Causation and effectuation: Toward a theoretical shift from economic inevitability to entrepreneurial contingency. *Academy of management Review* 26, 243-263 (2001)

The Problem Space for Effectuation

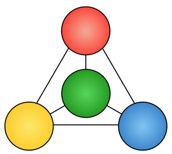
Where do we find rationality when

- *the environment does not independently influence outcomes or even rules of the game (Weick, 1979),*
- *the future is truly unpredictable (Knight, 1921), and*
- *the decision maker is unsure of his/ her own preferences (March, 1982)?*

	Existing Market	**New Market**
Existing Product		
New product		**Suicide Quadrant**

Knowledge Innovation
Maturity Dimensions

- *The application domain* (problem) *maturity dimension, which resonates with the key roles in innovation of opportunities, tasks and problems, markets, needs and fields.*



- *The knowledge* (solution) *maturity dimension, which resonates with the key roles in innovation of new ideas, new insights, new knowledge and skills, technological know-how, new knowledge, and learning.*

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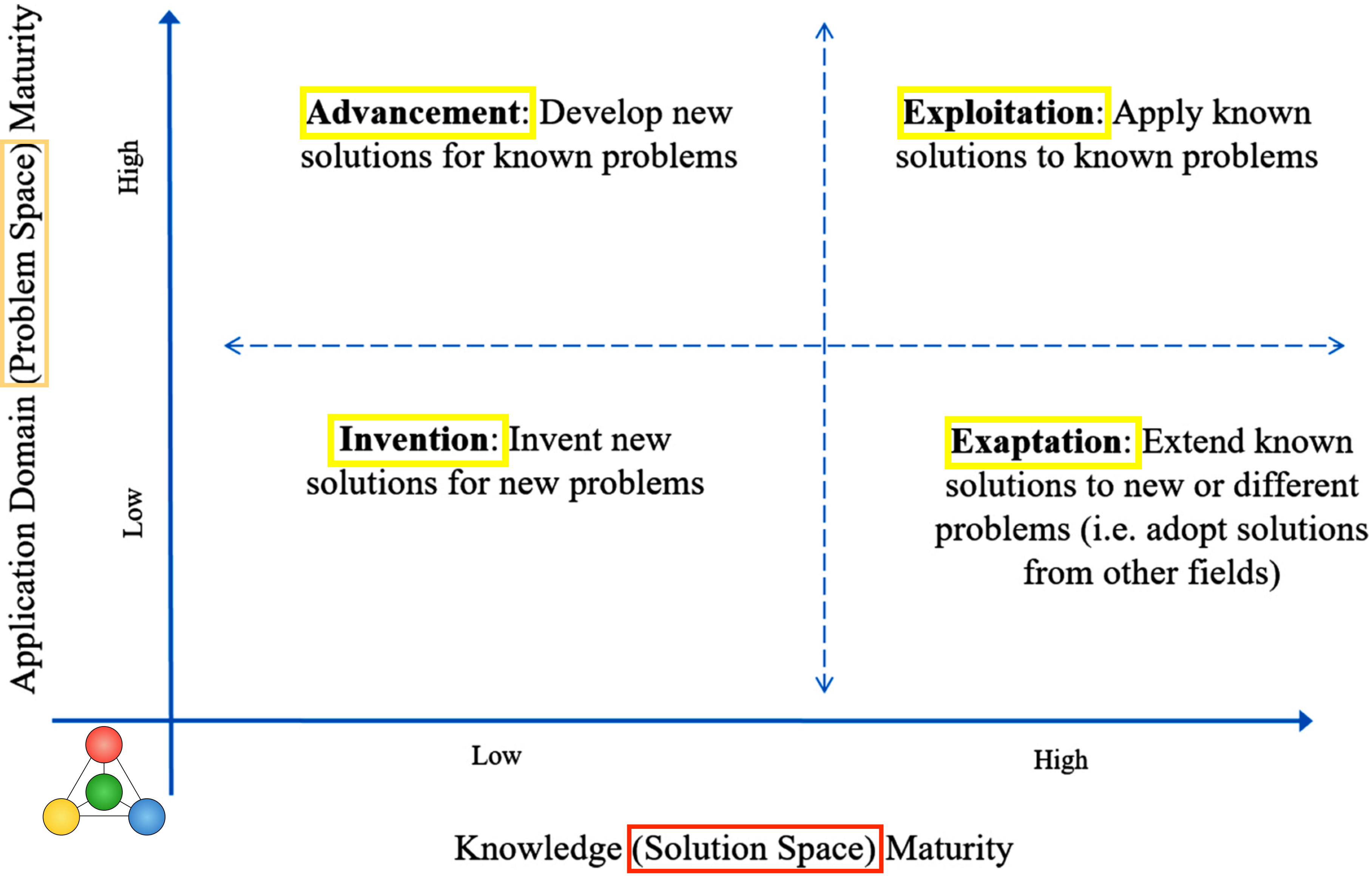
KIM: Knowledge Innovation Matrix



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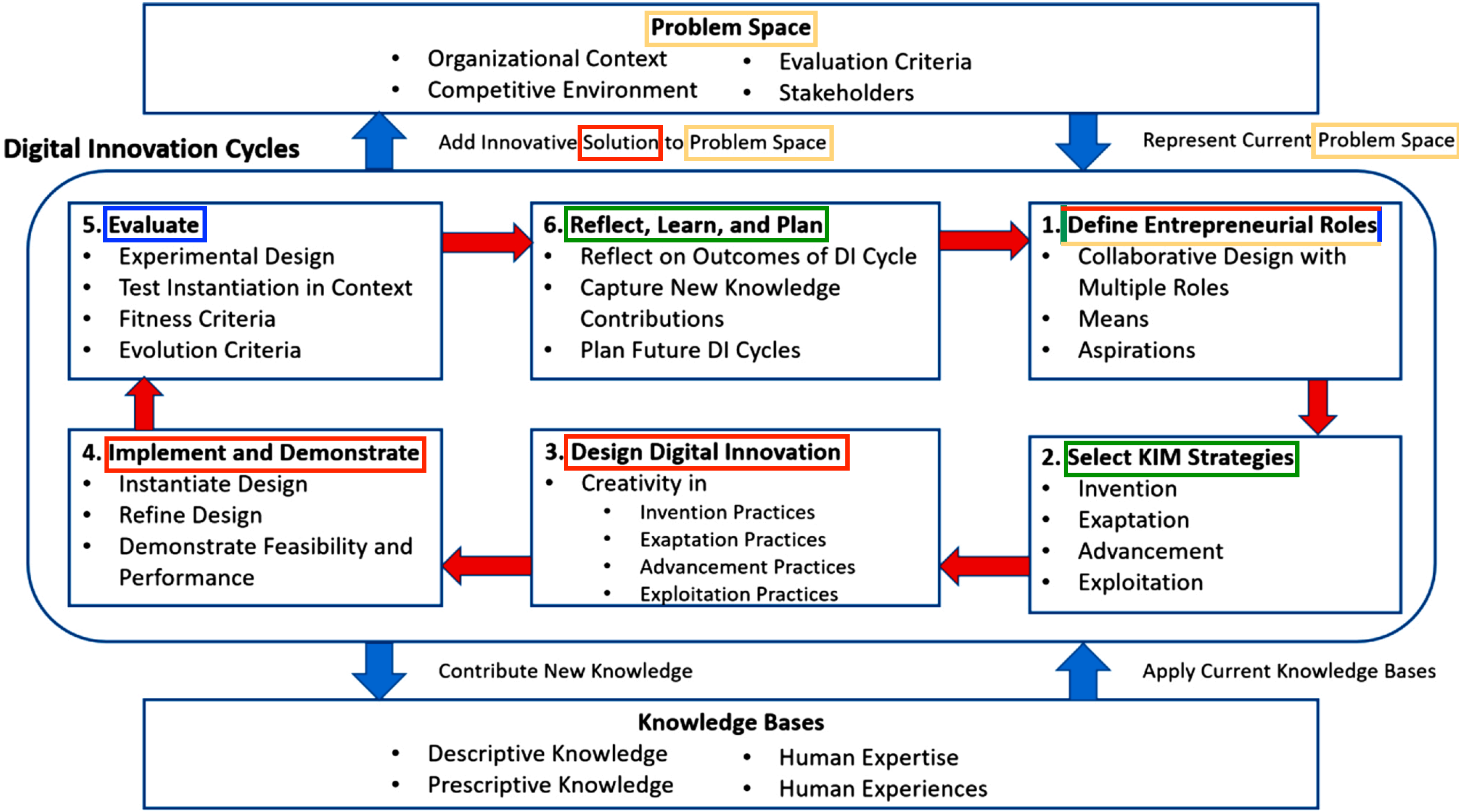
Process model with strategies and roles



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Discussion of the paper

- Focus on processes (rather than structures).
- Focus on strategy (as opposed to rationale or tactics).
- Catalog of strategies (as opposed to tools and techniques).
- Integrates innovation and entrepreneurship (as opposed to keeping them separate)
- Views invention as one form of innovation (as opposed to viewing innovation as the combination of invention, exploitation, and diffusion).
- Main focus on management (rather than development).
- Tends to favor effectuation with a hint of causation.